

GEN AI MASTERCLASS

Co-Pilots | RAG | Agents | MCP | Product Building

6-Week Intensive Cohort

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OXFORD UNIVERSITY GenAI TRAINER

SCHEDULE OVERVIEW

Weekend Sessions (Sat & Sun)	2:00 PM – 4:00 PM IST	Theory, Demos & Live Coding
Office Hours (Wednesday)	7:00 PM – 9:00 PM IST	Practice, Q&A & Assignment Review

Weekend sessions cover theory and demos. Wednesday office hours are for hands-on practice, questions, and assignment reviews.

WEEK-BY-WEEK CURRICULUM

WEEK 1: Foundations & LLM Essentials

Day 1 (Sat)	SDLC with LLM, Dev Tools & Tech, Project Scaffolding (basic website building), Introduction and App Ideas
Day 2 (Sun)	Introduction to LLM — architecture, capabilities, key concepts, and use cases
OFFICE HOURS:	Practice: Set up dev environment, scaffold first project, explore LLM APIs

WEEK 2: ChatBots & Prompt Engineering

Day 1 (Sat)	Building Basic ChatBot using LangChain, OpenAI API Key Integration
Day 2 (Sun)	Prompt Engineering — techniques, patterns, and real-life use cases
OFFICE HOURS:	Practice: Build and iterate on chatbot, prompt engineering exercises

WEEK 3: Vector DBs & RAG Foundations

Day 1 (Sat)	Vector DB Deep Dive (Qdrant) — embeddings, similarity search, indexing
Day 2 (Sun)	RAG Deep Dive — Building Chat Over Document with basic and Self RAG
OFFICE HOURS:	Practice: Implement vector search, build first RAG pipeline

GEN AI MASTERCLASS 6-Week Cohort — Weeks 4–6 & Details

WEEK-BY-WEEK CURRICULUM (CONTINUED)

WEEK 4: Advanced RAG & AI Agents

Day 1 (Sat)	RAG Deep Dive continued — Building Chat Over Document with advanced and Self RAG techniques
Day 2 (Sun)	AI Agents — introduction to autonomous agents, design patterns, and real-world applications
OFFICE HOURS:	Practice: Build advanced RAG system, explore agent architectures

WEEK 5: Agent Orchestration & MCP

Day 1 (Sat)	AI Agents continued with LangGraph — orchestration, state management, multi-step workflows
Day 2 (Sun)	MCP — Model Context Protocol, deploying modular AI tools as microservices
OFFICE HOURS:	Practice: Build multi-agent system, set up MCP server

WEEK 6: Advanced Applications & Capstone

Day 1 (Sat)	Advanced AI Application Implementation with AI Agents and MCP — full production workflow
Day 2 (Sun)	Offline model setup and training models — Capstone Project Review and Closure
OFFICE HOURS:	Final Q&A, project polish, certificate discussion

LEARNING OUTCOMES

- Build end-to-end AI systems using LangChain, LangGraph, and modern GenAI frameworks
- Implement RAG pipelines with vector databases (Qdrant) for document intelligence
- Design, orchestrate, and deploy autonomous AI agents for real-world applications
- Deploy AI microservices using MCP Server and cloud infrastructure (AWS, Docker)
- Master prompt engineering, LLM safety, guardrails, and production best practices

TECHNOLOGY STACK

Language:	Python
LLM Models:	OpenAI, DeepSeek, Claude
Frameworks:	LangChain, LangGraph, LangSmith
Vector Stores:	Qdrant DB, PG Vector
Graph DB:	Neo4j
Infrastructure:	AWS, Docker, MCP Server
Tracing:	Langfuse (Docker-based)

PREREQUISITES

- Cursor IDE — \$20/month subscription (or any preferred IDE)
- \$10–20 in OpenAI API Key Credit (or any other LLM provider)

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